Blockchain Based Information Security and Privacy Protection: Challenges and Future Directions using Computational Literature Review

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Abstract

Blockchain technology is an emerging digital innovation that has gained immense popularity in enhancing individual security and privacy within Information Systems (IS). This surge in interest is reflected in the exponential increase in research articles published on blockchain technology, highlighting its growing significance in the digital landscape. However, the rapid proliferation of published research presents significant challenges for manual analysis and synthesis due to the vast volume of information. The complexity and breadth of topics, combined with the inherent limitations of human data processing capabilities, make it difficult to comprehensively analyze and draw meaningful insights from the literature. To this end, we adopted the Computational Literature Review (CLR) to analyze pertinent literature's impact and topic modelling using the Latent Dirichlet Allocation (LDA) technique. We identified 10 topics related to security and privacy and provided a detailed description of each topic. From the critical analysis, we have observed several limitations, and several future directions are provided as an outcome of this review.

Keywords: Computational Literature Review, Blockchain, Latent Dirichlet Allocation, Security and Privacy, Information Systems.

1. Introduction

Blockchain, a distributed ledger system, has garnered significant attention from industry experts and academics as a potential solution for developing secure and privacy-oriented information systems (IS). The convergence of various cutting-edge technologies has brought new challenges to protecting information. security and users' privacy (Hughes et al., 2019). In today's world, data sharing and analyzing personal information are common practices in interdomain communication systems. Therefore, any security breach can lead to identity theft, confidentiality issues, financial losses, business damage, and loss of customer trust (Wylde et al., 2022). However, Blockchain integration addresses these issues as its decentralized and tamper-proof ledger provides enhanced security and privacy through cryptographic verification and consensus mechanisms (Puthal et al., 2018).

In recent years, significant advancements have been made in utilizing blockchain technology to enhance the privacy and security of IS. Researchers have developed various methodologies integrating advanced security and privacy protection protocols like Zero-Knowledge Proof (ZKP), Multiparty Computations (MPC), and Fully Homomorphic Encryption to address security and privacy issues on the blockchain (Ma et al., 2023; Waheed et al., 2020; Wang et al., 2023). The research on advanced security and privacy protection protocols has led to an increasing number of articles being published exclusively on IS, highlighting the challenges in developing security and privacy-oriented blockchains. To develop an enhanced version of blockchain architecture, it is essential to analyze the current literature (Homoliak et al., 2020), which helps to identify critical research gaps and formulate research questions that address pertinent issues related to security and privacy on the blockchain.

Furthermore, keeping up-to-date with this field's latest topics and trends is crucial for researchers to advance their work. Scholarly publications on blockchain-based privacy and security systems are significantly increasing. Researchers are doing systematic and bibliometric reviews of previous works to identify challenges, opportunities, and potential

solutions. However, biases arise when selecting articles from an extensive collection. To address this challenge, we use the CLR topic modeling method (Antons et al., 2023), a technique used to uncover the hidden thematic structure from a large corpus of literature. CLR enables the conduct of a more comprehensive and unbiased analysis of the literature of around 3,904 articles. This analysis helps us identify the top trending topic of privacy and security using blockchain in IS. The CLR helps us recognize critical future research questions for privacy-preserving blockchain architecture. We found the absence of CLR methods on blockchain during previous work reviews for security and privacy studies. This is the key motivation for our research, as we saw the potential for CLR to enhance the quality and objectivity of such reviews in the field of information security and privacy using blockchain for IS.

This article is structured as follows: First, we discuss CLR in detail about blockchain-related works in IS. Then, we explain the methods and tools used in this work, which include impact and content analysis to identify the topics. The next section provides analysis results with visualizations. Finally, we will discuss each topic in detail and describe it. Based on this discussion, we developed research questions.

2. Background

2.1 Computational Literature Review

There has been a significant increase in the large amount of literature, which challenges scholars in identifying specific research domains. Scholars use traditional literature review methods such as systematic and bibliometric literature reviews to minimize bias in article selection (Thilakaratne et al., 2019). Systematic literature reviews (SLRs) (Nightingale, 2009) have emerged as the preferred method for rigorous and reproducible literature analysis. Further, researchers find new techniques and tools that combine computational techniques with the robustness of SLR. This technique increases the number of performances and reduces the complexity of the review process. One such method is CLR (Antons et al., 2023), which uses advanced machine learning (ML) algorithms to analyze large amounts of textual data systematically. The automated process of CLR accelerates the review process and increases the scope of insights gained from the literature. CLR is based on topic modelling algorithms, such as Latent Dirichlet Allocation (LDA) (Blei et al., 2003), which identifies latent themes within a corpus of documents. The LDA topic modeling works on the unsupervised probabilistic statistical model to identify the hidden topics in the corpus from the content of the articles (Blei et al., 2003).

To perform CLR, researchers follow a structured framework involving six key steps:

1.Begin with a conceptual goal that is the researcher's motivation to perform the review.

2.Operationalising the CLR helps identify the boundaries for selecting the content and the area of the work.

3.Choose a computational technique regarding the suitability of the corpus of the document.

4.Perform the content analysis by preparing the data based on the selected computational algorithm in previous steps.

5.Generate original insights by organizing the evaluation of outputs in step four.

6.Present the findings by applying synthesis forms like research agenda, taxonomy, models, meta-analysis, and meta-theory to present the findings and synthesize the insights into useful building blocks for further research.

3. Previous Literature Works

In recent years, there has been a surge of interest in blockchain technology, leading to various research efforts in new domains, as observed by exploring databases such as Scopus and IEEE Explorer. Numerous systematic literature review (SLR) papers have been published on blockchain on security and privacy challenges. Mohanta et al. (2019) conducted an SLR on 150 articles to analyze the applications of blockchain technology, in which 20 articles discussed the challenges related to security and privacy in blockchain implementation. A study by Zhang et al. (2019) analyzed 14 articles to delve into blockchain security and privacy techniques and provides detailed descriptions of consensus algorithms, hash chained storage, mixing protocols, anonymous signatures, non-interactive ZKP, and secure MPC to achieve security and privacy in blockchain-based systems. Waheed et al. (2020) presented a technique that leverages ML to develop automated privacy and security mechanisms. Based on an analysis of 43 articles from journals and conferences, it evaluates security and privacy challenges and vulnerabilities in previous works. Ismagilova et al. (2022) reviewed 99 articles focusing on the privacy and security of mobile devices in smart cities, power systems, healthcare, and blockchain. Gugueoth et al. (2023) reviewed 31 articles on IoT, describing types of cyber attacks on IoT systems that can lead to privacy and security

vulnerabilities and also explored solutions using blockchain to solve these vulnerabilities and improve IoT architecture. The SLR performed on 51 articles by Kiania et al. (2023) on the benefits of adopting blockchain in healthcare systems and its role in improving security and privacy. It also discussed the limitations of blockchain adoption in healthcare systems. A study presented by Qahtan et al. (2023) provides a comprehensive taxonomy on attribute-related security and privacy developments for blockchain-based healthcare industry 4.0. This study focused on identifying issues in healthcare industry 4.0 and addressing challenges in multi-criteria evaluation, proposing specific methods to bridge theoretical gaps. Myrzashova et al. (2023) presents an SLR with 100 articles that show how the advancement of privacy in healthcare led to the adoption of Federated Learning (FL) in healthcare systems. This study's main focus is identifying vulnerabilities in user privacy protection systems and healthcare data. Against that, the authors presented a novel conceptual framework for blockchain-enabled FL

Table 1. Comparison with existing literature review studies

Criteria	(Mohanta et al., 2019)	(Zhang et al., 2019)	(Waheed et al., 2020)	(Myrzashova et al., 2023)	(Qahtan et al., 2023)	(Gugueoth et al., 2023)	(Shahid, 2020)	Our
1	20	14	43	100	52	31	2125	3904
2	×	×	✓	✓	✓	✓	✓	✓
3	✓	✓	✓	✓	✓	✓	✓	✓
4	×	×	×	×	×	×	✓	✓
5	×	×	×	×	×	×	✓	✓
6	×	×	×	×	×	✓	✓	✓
7	×	×	×	×	×	×	✓	✓
8	×	×	×	×	×	×	✓	✓
9	✓	✓	✓	✓	✓	✓	✓	✓
10	×	×	×	✓	×	×	✓	✓
11	✓	×	×	✓	✓	✓	✓	✓
12	×	×	×	×	×	×	✓	✓

Note, numeric values indicate as follows: 1= number of literature included in the analysis, 2 = Quantitative Analysis, 3 = Qualitative Analysis, 4 = Data Collection Automation, 5 = Scalability, 6 = Real-Time Analysis, 7 = Algorithmic Topic Evolution Tracking, 8 = Dynamic Literature Synthesis, 9 = Trends and Pattern Identification, 10 = Unstructured Data Processing, 11 = Data-Driven Insights, 12 = Latent Topic Insight.

The literature mentioned above utilizes systematic and comprehensive literature reviews. However, it has been criticized for its inability to effectively synthesize content when dealing with a large amount of academic literature. Manual techniques and citation analysis become impractical in such cases. Shahid (2020) introduced a topic modeling-based literature review process to address this issue, using a corpus of 2,125 entries. The study employed LDA to identify topics within the corpus. The research covered practical and academic domains, with computer science being the dominant research area, followed by economics and business. This underscores the necessity for research on blockchain development across various disciplines and suggests potential future research directions. Building on this, we have adopted an unsupervised topic modeling approach to uncover trends specifically related to the security and privacy of blockchain systems. Table 1 presents the comparison between our work and previous literature work.

3. Methodology

This study aims to identify emerging fields and promising paths for future research related to blockchain applications for individual users and organizations. Our methodology follows a step-by-step approach.

3.1. Data Collection

We started our data collections with searching articles on different databases such as IEEE, Web of Science, and Scopus, using keywords with search logic as "Blockchain" AND "Security" AND "Privacy" AND "Decentrali*" OR "Distributed". The search outcomes overlap with different databases, and most of the relevant articles are indexed in Scopus. It is also found that the collection of Scopus is comprehensive, multidisciplinary, and trusted based on abstract and citation parameters. Therefore, the Scopus database was used for conducting the CLR method through which we collected 4,975 relevant articles.

After careful filtering by subject area, article type, and language, we finally selected 4,016 articles, which we exported as a CSV file. These articles were limited to journal and conference publications in English with subject tags in Computer Science Engineering, Mathematics, Decision Sciences, Energy, Medicine, Business, Management and Accounting, Health Professions, Social Sciences, Environmental Science, Economics, Econometrics, and Finance. We